

Homework 7

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5.3.2

$$\begin{aligned}
 \vec{j} &= \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) \\
 &= \frac{\hbar}{2mi} [c^* \tilde{\psi}^* \nabla (c \tilde{\psi}) - c \tilde{\psi} \nabla (c^* \tilde{\psi}^*)] \quad \leftarrow \psi = c \tilde{\psi} \\
 &= \frac{\hbar}{2mi} (c^* c \tilde{\psi} \nabla \tilde{\psi} - c c^* \tilde{\psi} \nabla \tilde{\psi}) \quad \leftarrow \tilde{\psi} \text{ is real} \\
 &= \frac{\hbar}{2mi} |c|^2 (\tilde{\psi} \nabla \tilde{\psi} - \tilde{\psi} \nabla \tilde{\psi}) \\
 &= 0
 \end{aligned}$$

5.3.3

* Formula: $\nabla e^{i(\vec{p} \cdot \vec{r})/\hbar} = \frac{i\vec{p}}{\hbar} e^{i(\vec{p} \cdot \vec{r})/\hbar}$

$$\begin{aligned}
 \nabla e^{i(\vec{p} \cdot \vec{r})/\hbar} &= \nabla e^{i(p_x x + p_y y + p_z z)/\hbar} \\
 &= \hat{e}_x \frac{\partial}{\partial x} e^{i(p_x x + p_y y + p_z z)/\hbar} + \hat{e}_y \frac{\partial}{\partial y} e^{i(p_x x + p_y y + p_z z)/\hbar} + \hat{e}_z \frac{\partial}{\partial z} e^{i(p_x x + p_y y + p_z z)/\hbar} \\
 &= \left(\frac{i}{\hbar} p_x \hat{e}_x + \frac{i}{\hbar} p_y \hat{e}_y + \frac{i}{\hbar} p_z \hat{e}_z \right) e^{i(p_x x + p_y y + p_z z)/\hbar} \\
 &= \frac{i\vec{p}}{\hbar} e^{i(\vec{p} \cdot \vec{r})/\hbar}
 \end{aligned}$$

$$\begin{aligned}
 \vec{j} &= \frac{\hbar}{2mi} (\psi_p^* \nabla \psi_p - \psi_p \nabla \psi_p^*) \\
 &= \frac{\hbar}{2mi} \left[\left(\frac{1}{2\pi\hbar} \right)^{\frac{3}{2}} e^{-i(\vec{p} \cdot \vec{r})/\hbar} \nabla \left(\frac{1}{2\pi\hbar} \right)^{\frac{3}{2}} e^{i(\vec{p} \cdot \vec{r})/\hbar} - \left(\frac{1}{2\pi\hbar} \right)^{\frac{3}{2}} e^{i(\vec{p} \cdot \vec{r})/\hbar} \nabla \left(\frac{1}{2\pi\hbar} \right)^{\frac{3}{2}} e^{-i(\vec{p} \cdot \vec{r})/\hbar} \right] \\
 &= \frac{\hbar}{2mi} \left[\left(\frac{1}{2\pi\hbar} \right)^3 e^{-i(\vec{p} \cdot \vec{r})/\hbar} \cdot \frac{i\vec{p}}{\hbar} e^{i(\vec{p} \cdot \vec{r})/\hbar} - \left(\frac{1}{2\pi\hbar} \right)^3 e^{i(\vec{p} \cdot \vec{r})/\hbar} \left(-\frac{i\vec{p}}{\hbar} \right) e^{-i(\vec{p} \cdot \vec{r})/\hbar} \right] \\
 &= \frac{\hbar}{2mi} \cdot \left(\frac{1}{2\pi\hbar} \right)^3 \cdot 2 \frac{i\vec{p}}{\hbar} \\
 &= \left(\frac{1}{2\pi\hbar} \right)^3 \frac{\vec{p}}{m}
 \end{aligned}$$

$$P = \psi_p^* \psi_p = \left(\frac{1}{2\pi\hbar} \right)^3$$

Therefore, $\vec{j} = P \frac{\vec{p}}{m}$. This is equivalent to $\vec{j} = P \vec{v}$, where $P \leftrightarrow \rho$, and $\frac{\vec{p}}{m} \leftrightarrow \vec{v}$. Since $P, \vec{p}$ are constant, $\frac{\partial P}{\partial t} = 0$, $\nabla \cdot \vec{j} = \frac{P}{m} \nabla \cdot \vec{p} = 0$. The continuity equation is trivial: $\frac{\partial P}{\partial t} = -\nabla \cdot \vec{j}$.

5.3.4

$$\begin{aligned}
& \psi^* \frac{\partial}{\partial x} \psi - \psi \frac{\partial}{\partial x} \psi^* \\
&= (A^* e^{-ipx/\hbar} + B^* e^{ipx/\hbar}) \left(\frac{ip}{\hbar} A e^{ipx/\hbar} - \frac{ip}{\hbar} B e^{-ipx/\hbar} \right) - \\
& \quad (A e^{ipx/\hbar} + B e^{-ipx/\hbar}) \left(-\frac{ip}{\hbar} A^* e^{-ipx/\hbar} + \frac{ip}{\hbar} B^* e^{ipx/\hbar} \right) \\
&= \frac{ip}{\hbar} |A|^2 - \frac{ip}{\hbar} |B|^2 + \frac{ip}{\hbar} AB^* e^{2ipx/\hbar} - \frac{ip}{\hbar} A^* B e^{-2ipx/\hbar} - \left(-\frac{ip}{\hbar} |A|^2 + \frac{ip}{\hbar} |B|^2 \right) \\
& \quad - \frac{ip}{\hbar} A^* B e^{-2ipx/\hbar} + \frac{ip}{\hbar} AB^* e^{2ipx/\hbar} \\
&= \frac{2ip}{\hbar} (|A|^2 - |B|^2)
\end{aligned}$$

Therefore,

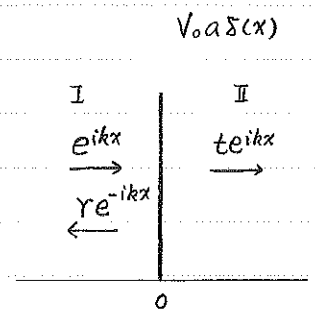
$$\begin{aligned}
j &= \frac{\hbar}{2mi} (\psi^* \frac{\partial}{\partial x} \psi - \psi \frac{\partial}{\partial x} \psi^*) \\
&= \frac{\hbar}{2mi} \cdot \frac{2ip}{\hbar} (|A|^2 - |B|^2) \\
&= \frac{p}{m} (|A|^2 - |B|^2)
\end{aligned}$$

5.4.2

* Recall that the boundary conditions for a δ -potential are: (see Homework 6 5.2.3)

$$\begin{cases} \psi_I(0) = \psi_{II}(0) \end{cases} \quad (1)$$

$$\begin{cases} \psi'_{II}(0) - \psi'_I(0) = \frac{2m}{\hbar^2} V_0 a \psi(0) \end{cases} \quad (2)$$

(a) For $x \neq 0$, we have

$$\psi''(x) + \frac{2m}{\hbar^2} E \psi(x) = 0$$

$$\text{Set } k = \sqrt{\frac{2m}{\hbar^2} E}$$

For this problem, we can set

$$\begin{cases} \psi_I(x) = e^{ikx} + r e^{-ikx} \end{cases} \quad (3)$$

$$\begin{cases} \psi_{II}(x) = t e^{ikx} \end{cases} \quad (4)$$

Substitute (3) and (4) into (1) and (2), we have

$$\begin{aligned}
& \begin{cases} 1 + r = t \\ ikt - ik + ikr = \frac{2m}{\hbar^2} V_0 a t \end{cases} \\
\Rightarrow & \begin{cases} t = \frac{1}{1 + \frac{imV_0 a}{k\hbar^2}} \\ r = \frac{-\frac{imV_0 a}{k\hbar^2}}{1 + \frac{imV_0 a}{k\hbar^2}} \end{cases}
\end{aligned}$$

$$R = |r|^2 = \frac{\frac{m^2 V_0^2 a^2}{k^2 \hbar^4}}{1 + \frac{m^2 V_0^2 a^2}{k^2 \hbar^4}} = \frac{1}{1 + \frac{2\hbar^2 E}{m a^2 V_0^2}}$$

$$T = 1 - R = \frac{1}{1 + \frac{m a^2 V_0^2}{2\hbar^2 E}}$$

(b)

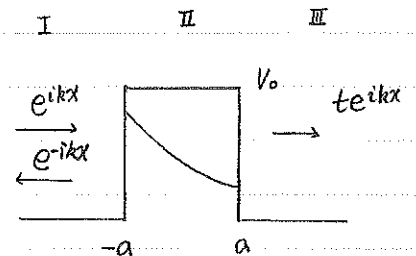
$$\text{I, III: } \psi''(x) + \frac{2m}{\hbar^2} E \psi(x) = 0$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

Set

$$\psi_{\text{I}}(x) = e^{ikx} + r e^{-ikx}$$

$$\psi_{\text{III}}(x) = t e^{ikx}$$



$$\text{II: } \psi''(x) + \frac{2m}{\hbar^2} (E - V_0) \psi(x) = 0$$

$$k = \sqrt{\frac{2m}{\hbar^2} (V_0 - E)}$$

$$\psi_{\text{II}} = A e^{kx} + B e^{-kx}$$

Boundary Conditions:

$$\begin{cases} \psi_{\text{I}}(-a) = \psi_{\text{II}}(-a) \\ \psi'_{\text{I}}(-a) = \psi'_{\text{II}}(-a) \end{cases} \Rightarrow \begin{cases} e^{-ika} + r e^{ika} = A e^{-ka} + B e^{ka} \\ i k e^{-ika} - i k r e^{ika} = k A e^{-ka} - k B e^{ka} \end{cases}$$

$$\begin{cases} \psi_{\text{II}}(a) = \psi_{\text{III}}(a) \\ \psi'_{\text{II}}(a) = \psi'_{\text{III}}(a) \end{cases} \Rightarrow \begin{cases} t e^{ika} = A e^{ka} + B e^{-ka} \\ i k t e^{ika} = k A e^{ka} - k B e^{-ka} \end{cases}$$

$$\Rightarrow t = \frac{4 e^{-2ika}}{4 \cosh 2ka + \left(\frac{k}{ik} + \frac{ik}{k}\right) \cdot 2 \sinh 2ka}$$

$$T = |t|^2 = \frac{1}{1 + \frac{1}{4} \sinh^2 2ka \cdot \left(\frac{k^2 + k^2}{kk}\right)^2}$$

$$R = 1 - T = \frac{\frac{1}{4} \sinh^2 2ka \cdot \left(\frac{k^2 + k^2}{kk}\right)^2}{1 + \frac{1}{4} \sinh^2 2ka \cdot \left(\frac{k^2 + k^2}{kk}\right)^2}$$

Problem 3

$$\Delta X(t) = \frac{\Delta(t)}{\sqrt{2}} = \frac{\Delta}{\sqrt{2}} \left(1 + \frac{\hbar^2 t^2}{m^2 \Delta^4} \right)^{\frac{1}{2}} \quad (\text{Shankar 5.1.17})$$

$$\Rightarrow t = \left[\left(\frac{\sqrt{2} \Delta X}{\Delta} \right)^2 - 1 \right]^{\frac{1}{2}} \left(\frac{m^2 \Delta^4}{\hbar^2} \right)^{\frac{1}{2}}$$

Substitute $m = 0.6 \text{ kg}$, $\Delta = \sqrt{2} \cdot 0.01 \text{ cm}$, $\Delta X = 100 \text{ m}$, we can get $t \sim 10^{32} \text{ s}$.